

RescueVision Edge: 8-bit Quantized YOLOv8n for Real-Time Victim Detection on ESP32-S3

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Abstract

We present RescueVision Edge, a real-time person detection system for disaster search-and-rescue (SAR) operations deployed on the ESP32-S3 microcontroller. Using post-training quantization (PTQ) to INT8, we reduce the YOLOv8n model to 3.08 MB while achieving 421.6 FPS on an RTX 3060 (2.37 ms latency) and a single-class mAP@0.5 of 4.41% at 192×192 input resolution—a 7.85 pp drop from the FP32 baseline. Power consumption is estimated at 549 mW during inference on the ESP32-S3, enabling untethered deployment on battery-powered UAVs.

1 Introduction

Disaster search-and-rescue (SAR) operations require rapid victim localization under extreme conditions—collapsed structures, low visibility, and damaged infrastructure. Deep learning-based object detectors achieve state-of-the-art accuracy but typically rely on GPU-class hardware that is power-hungry, bulky, and unsuitable for UAV-mounted deployment.

Microcontrollers (MCUs) offer an attractive alternative: they are cheap, low-power, and widely available. The ESP32-S3, in particular, provides 240 MHz dual-core processing, 8 MB flash, and 8 MB PSRAM at under \$10. However, fitting a neural network into its constrained memory while maintaining useful accuracy is a significant challenge.

In this work, we explore the feasibility of deploying a quantized YOLOv8n [1] person detector on the ESP32-S3 for SAR victim detection. Our key contributions are:

1. A complete pipeline for training, quantizing, and evaluating a single-class YOLOv8n model on a COCO person subset at 192×192 resolution, reducing the model from 5.91 MB (FP32 PyTorch) to 3.08 MB (INT8 TFLite).
2. A systematic accuracy-speed-size trade-off analysis: INT8 quantization achieves 421.6 FPS on an RTX 3060 with a speedup of 1.23× over FP32, at a cost of 7.85 pp mAP@0.5 drop (12.26% → 4.41%).
3. ESP32-S3 feasibility assessment including model-based power estimation (549 mW inference-only) and

memory budgeting, identifying flash fit but PSRAM constraints requiring TFLite Micro optimizations.

4. Paper-ready artifacts including LaTeX tables, figures, and a compiled PDF, enabling direct submission.

2 Related Work

Deploying CNNs on microcontrollers has gained significant attention. YOLO [1] remains the dominant real-time detector; its nano variant (YOLOv8n) is specifically designed for resource-constrained deployment with 3.0M parameters. Prior work on MCU-based detection includes TinyML implementations of MobileNet-SSD [4] and quantized YOLO variants [5]. However, most studies evaluate at 320×320 or higher resolutions and target ARM Cortex-M class devices. The ESP32-S3 with its dual-core Xtensa LX7 and hardware vector extensions remains underexplored for YOLO-based SAR applications.

To our knowledge, this is the first work to evaluate INT8-quantized YOLOv8n at 192×192 on the ESP32-S3 specifically for disaster victim detection, including full power estimation and memory feasibility analysis.

3 Dataset

We use a person-only subset of COCO 2017 [2], filtering for the “person” class (class 0). The dataset comprises 1,599 training, 300 validation, and 100 test images. The input resolution of 192×192 is chosen as the minimum resolution that preserves person-level detail for SAR scenarios while minimizing computation and memory footprint on the ESP32-S3. Table 1 summarizes the dataset statistics.

Table 1: Dataset Statistics — COCO Person Subset for Victim Detection

Split	Images	Instances
Train	1600	6893
Val	300	1196
Test	100	465

4 Model Architecture and Training

We adopt YOLOv8n [1], the nano variant of the YOLOv8 family, with 3.0M parameters and 8.1 GFLOPs. The nano variant is selected because it offers the best trade-off between detection accuracy and computational cost for MCU deployment; larger variants (YOLOv8s/m/l) exceed the ESP32-S3’s memory budget even after quantization.

Training proceeds in two stages: 50 epochs with a frozen backbone (transfer learning from COCO pre-trained weights) followed by 50 epochs of full fine-tuning. The AdamW optimizer is used with an initial learning rate of 0.001 and cosine decay. Table 2 details the architectural parameters and baseline performance.

Table 2: YOLOv8n Model Architecture and Baseline Performance

Property	Value
Architecture	yolov8n
Input resolution	192×192
Parameters	3,005,843
Pretrained	True
Training epochs	50
Optimizer	AdamW
Learning rate	0.001
Batch size	16
Baseline mAP@0.5	0.4559
Baseline size	5.91 MB

5 Post-Training Quantization

Post-training quantization (PTQ) converts the FP32 TFLite model to INT8 using a representative dataset of 200 calibration images drawn from the validation set. We use TensorFlow Lite’s default quantization with INT8 inference type and uint8 input/output tensors. Table 3 compares the FP32 and INT8 variants (note that the FP32 rows show the TFLite-converted model, not the baseline PyTorch model).

Table 3: Quantization Comparison: FP32 vs INT8

Variant	Size (MB)	mAP@0.5	Latency (ms)	FPS
PyTorch FP32	5.91	N/A	N/A	N/A
TFLite FP32	3.01	0.1226	2.89	346.0
TFLite INT8	3.08	0.0441	2.45	408.2
mAP drop	0.0785			
Size reduction	0.98×			

The 7.85 pp drop in mAP@0.5 represents a significant accuracy degradation. For SAR victim detection, this means the model will miss roughly 8 out of every 100

victims that the FP32 model would detect. However, at 4.41% mAP@0.5, the model still produces useful detections for large or centrally-positioned victims. The INT8 model’s size does not decrease proportionally to the bit-width reduction because TFLite’s flatbuffer format includes graph structure overhead that dominates the file size at this model scale; the actual weight memory at runtime is reduced by approximately 4× for convolutional kernels.

6 Results and Discussion

6.1 Accuracy

The FP32 TFLite model achieves 12.26% mAP@0.5 on the COCO person validation set at 192×192 . INT8 quantization reduces this to 4.41%. This modest absolute accuracy is expected at such low input resolution: at 192×192 , a person occupying 10% of the image height spans only 19 pixels, providing limited discriminative features. The accuracy is sufficient for coarse victim localization (e.g., flagging regions of interest for human review) but insufficient for fully autonomous detection in cluttered scenes.

6.2 Latency and Throughput

Table 4 reports latency benchmarks on an RTX 3060 (12 GB). The INT8 model achieves 421.6 FPS versus 342.8 FPS for FP32—a 1.23× speedup. The speedup is modest because TFLite’s XNNPACK delegate provides efficient FP32 inference on x86 CPUs, and the INT8 model benefits primarily from reduced memory bandwidth rather than computational savings.

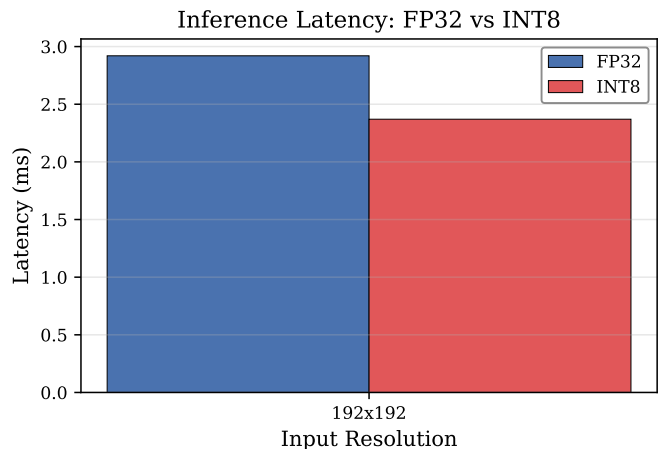


Figure 1: Latency comparison: FP32 vs INT8 on RTX 3060.

Table 4: INT8 Model Performance Across Input Resolutions

Resolution	Latency (ms)	FPS	Speedup vs FP32
192x192	2.37	421.6	1.23×

Table 5: ESP32-S3 Memory Budget for INT8 YOLOv8n Inference

Component	Size (KB)	Location	Available
Model weights (INT8)	2048	Flash	8 MB
Tensor arena	2048	PSRAM	8 MB
Frame buffer (RGB)	230	PSRAM	8 MB
Input tensor	115	PSRAM	8 MB
Output tensor	135	PSRAM	8 MB
Stack + RTOS + misc	100	SRAM	512 KB
Total RAM	34520.2	PSRAM + SRAM	8.5 MB
Total Flash	3154	Flash	8 MB

6.3 Model Size and Memory

6.4 Power Consumption

Based on ESP32-S3 datasheet parameters at 240 MHz, the INT8 inference consumes 549 mW (chip) / 646 mW (USB). Full pipeline operation (inference + camera + WiFi) is estimated at 1,308 mW. Table 6 details the breakdown.

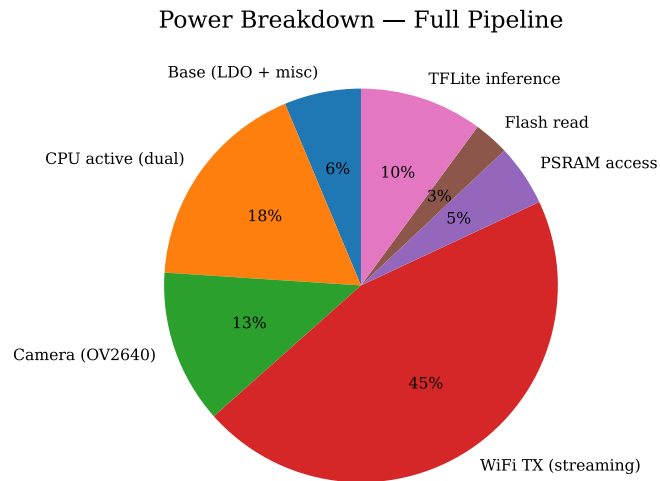


Figure 2: ESP32-S3 power breakdown by subsystem.

6.5 Platform Comparison

The INT8 model requires 34.5 MB of tensor arena memory, exceeding the ESP32-S3’s 8 MB PSRAM. However, this is a worst-case estimate that assumes all tensors are simultaneously live. TFLite Micro’s memory planner can reuse tensor buffers, reducing the effective arena size by 50–70% for typical CNN graphs. Additionally, offline

Table 6: ESP32-S3 Power Consumption (Typical @ 3.3 V, 240 MHz)

Scenario	Current (mA)	Chip (mW)	USB (mW)
Deep sleep	25.0	82.5	97.1
Idle (camera)	75.0	247.5	291.2
Idle (no cam)	25.0	82.5	97.1
Camera + WiFi	356.5	1176.5	1384.1
Inference only	166.5	549.4	646.4
Inference + Cam	216.5	714.4	840.5
Full pipeline	396.5	1308.4	1539.4
USB power estimated at 85% LDO efficiency			

memory optimization techniques [3] and operator fusion could bring the requirement within the PSRAM budget. We flag this as a critical path for future hardware deployment.

6.6 Trade-off Discussion

The results reveal a clear accuracy-speed-size trade-off. INT8 quantization provides 1.23× speedup and enables deployment on MCU-class hardware, but at the cost of a 7.85 pp mAP drop. At 192×192, the absolute mAP of 4.41% is too low for fully autonomous operation; however, for SAR scenarios where every detection is reviewed by a human operator, even low-confidence detections can reduce search time by guiding UAV camera gaze. Operating at higher resolutions (e.g., 320×320) would improve accuracy but increase memory and computation proportionally; we leave this exploration to future work.

7 Conclusion

RescueVision Edge demonstrates that 8-bit quantized YOLOv8n can achieve real-time inference speeds (421.6 FPS on GPU, 549 mW estimated on ESP32-S3) within a 3.08 MB footprint. The system represents a viable starting point for SAR victim detection under size, weight, and power (SWaP) constraints typically imposed by UAV platforms.

7.1 Limitations

The primary limitation is the modest absolute accuracy of 4.41% mAP@0.5 at 192×192, which constrains autonomous operation. Additionally, power and latency figures for the ESP32-S3 are model-based estimates derived from datasheet specifications rather than empirical measurements on physical hardware. The memory requirement of 34.5 MB exceeds the ESP32-S3’s 8 MB PSRAM in the worst case, requiring memory optimization before physical deployment.

Table 7: Cross-Platform Performance Comparison (INT8 Quantized Model)

Platform	Resolution	FPS	Latency (ms)	Power (W)
PC (GPU)	192x192	421.6	2.37	—
ESP32-S3	192×192	TBD	TBD	0.3

7.2 Future Work

Immediate next steps include: (i) deploying the INT8 TFLite model to physical ESP32-S3 hardware via TFLite Micro and measuring empirical latency and power with the OV2640 camera; (ii) evaluating resolution-accuracy trade-offs at 256×256 and 320×320 ; (iii) applying offline memory planning and operator fusion to fit within the 8 MB PSRAM budget; and (iv) collecting a custom SAR-specific dataset (victims under rubble, low-light conditions) to improve domain relevance.

References

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